

Ethan Zhang

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EDUCATION

University of California, Berkeley

August 2024 - May 2028

B.A. Computer Science & B.A Applied Mathematics

Berkeley, CA

- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Cloud Computing, Distributed Systems

PROFESSIONAL EXPERIENCE

Workday

Pleasanton, CA

Software Engineer Intern

May 2026 - Present

- Gave **real-time accuracy visibility** into a production export pipeline processing **10M+ exports over 3 months** by instrumenting **Micrometer** metrics comparing predicted vs. actual duration on a live monitoring dashboard.
- **Suppressed 9% of estimates as low-confidence** before reaching customers by building an offline evaluation pipeline that computed **mean absolute error** across historical runs, root-causing low-data predictions instead of shipping them blind.
- **Coordinated with on-call engineers** to roll out the feature, benchmarking latency **under 20ms**, and authoring a runbook.

Toyota — Contract

Berkeley, CA

Software Engineer Consultant

January 2026 - May 2026

- **Unified staff-scheduling data** from 3 disparate sources into a normalized **PostgreSQL schema** covering **~200 shifts/week**.
- **Cut manual scheduling time from ~6 hours/week to under 1 hour** by designing and implementing a **greedy heuristic optimization algorithm** that automatically minimized scheduling conflicts and understaffing across the workforce.
- **Cut understaffed shifts 25%** versus the client's manual baseline by validating algorithm output against **8 weeks of historical scheduling data**, triaging and root-causing the edge cases the heuristic had initially mishandled.

Blueberry AI

Burlingame, CA

Software Engineer Intern

October 2025 - December 2025

- **Delivered real-time retrieval, processing, and rendering** for **100+ 3D model formats** for Blueberry's browser-based 3D asset viewer by developing **C++ parsing modules** integrated with RESTful APIs, AWS S3, and Node.js microservices.
- **Cut query latency 40%** across **4 million design assets** for 1K+ creative-team customers by building scalable, containerized backend systems (**Docker, Kubernetes, MongoDB**) integrating OpenAI and CLIP models to automate AI tagging.
- **Reduced production release errors 25%** and **increased test coverage to over 90%** via streamlined CI/CD pipelines.

Lumo

San Francisco, CA

Founding Software Engineer

March 2025 - Present

- **Architected a CMMS-agnostic ELT data platform** enabling new systems to onboard as isolated transforms.
- **Balanced analytical utility against data-sensitivity requirements** by designing a **data anonymization layer** stripping PII before it reaches the LLM, and hardening the system with **prompt-injection mitigation and LDAP/AD access control**.
- **Identified \$1.2M in annual labor inefficiency** and secured a **signed Letter of Intent for pilot deployment** by leading a 6-week enterprise proof-of-concept with Applied Materials (ABM), a semiconductor equipment manufacturer.

PROJECTS

Equivalence Systems, LLC

Berkeley, CA

AI Researcher

August 2025 - December 2025

- **Cut manual review time 90%** across **36M+ academic records** for university stakeholders by deploying a distributed NLP pipeline (**Python, HuggingFace, GCP**) with built-in preprocessing, validation, and continuous reliability monitoring.
- **Enabled sub-5-second processing of 10K+ transcripts** by building a parallelized OCR-to-JSON parser (**Tesseract, Pandas**) as a reusable CLI tool with structured logging and error handling to speed root-cause triage of failures.

SMCCD Intercollegiate Hackathon — 1st Place, Machine Learning Track

San Mateo, CA

Full Stack Engineer

April 2025

- **Won 1st place among 100+ teams:** built a real-time music recommender processing **3M+ plays** with sub-second personalized suggestions.
- **Raised precision to MAP@20 = 0.89 (+34%)** tuning collaborative filtering algorithms and optimizing API response paths with caching.

SKILLS

Languages & Frameworks: C++, Go, Java, Python, C, TypeScript, JavaScript (ES6+), React, Node.js, Flask, FastAPI, SQL

Cloud & Infrastructure: AWS (EC2, S3, Lambda), GCP (Firebase, Cloud Run), Docker, CI/CD Pipelines (GitHub Actions), Git, Linux

Databases & Systems: PostgreSQL, MySQL, Redis, API Architecture, Distributed Systems, Real-Time & Embedded Systems

Debugging & Observability: Monitoring & Metrics (Micrometer), Root-Cause Analysis & Triage, Incident Runbooks & On-Call Tooling